ITAI 1378 COMPUTER VISION & AI • MIDTERM BLUEPRINT

Project KADERO

Kinetic Automated Detection of Equipment in Real-time Operations

STUDENT NAME
Kamron Robinson

INSTITUTION
Houston Community College

PROJECT TIER
Tier 2 (Custom CNN + Multi-Object Tracking)

SPEAKER SCRIPT (~30 SEC)

"Hello everyone, I'm Kamron Robinson. Today I'm presenting the Blueprint for my midterm project, KADERO: Kinetic Automated Detection of Equipment in Real-time Operations. I'm building a Tier 2 computer vision system combining a custom PyTorch detector with object tracking to monitor worker PPE compliance across real-time video feeds."

PROBLEM LANDSCAPE

Worksite Safety Blind Spots & Compliance Risks

The Problem

Construction and industrial sites face severe injury hazards and strict OSHA fines when workers fail to wear mandatory safety gear like hard hats, vests, and gloves.

Who Cares?

Site safety managers, general contractors, EHS compliance inspectors, and facility operations directors responsible for zero-injury protocols.

Why Now?

Manual inspections are sporadic, subjective, and leave unmonitored blind spots. Continuous computer vision monitoring ensures safety protocols are upheld automatically.

SPEAKER SCRIPT (~40 SEC)

"Manual job-site safety checks cannot cover every angle at all times. When a worker removes their helmet, vest, or gloves in an active zone, the danger of preventable injury spikes immediately. KADERO automates site monitoring so safety supervisors can catch non-compliance instantly rather than after an accident happens."

SYSTEM PIPELINE

End-to-End Automated Compliance Pipeline

Core Solution: An automated computer vision pipeline combining custom object detection with multi-object tracking to detect PPE violations and track individual worker compliance over time.

STAGE 1
Video / Image Input
OpenCV capture & upload

→

STAGE 2
Custom CNN Detector
PyTorch boxes & classes

→

STAGE 3
Tracking Module
Persistent worker ID track

→

STAGE 4
Compliance & Alerts
Live status indicator

- **Compliant: Worker tracked with verified gear (green indicator).**
- **Violation: Worker tracked without required equipment or flagged with NO-Gloves (red indicator & logged alert).**

SPEAKER SCRIPT (~45 SEC)

"KADERO ingests live webcam video or image files, feeds each frame into a custom PyTorch detector to identify people and safety gear, and passes those detections to a tracking module to monitor individual worker status over time."

ENGINEERING ARCHITECTURE

Custom PyTorch Model & Tracking Stack

Detection Model

Custom Convolutional Neural Network (CNN) built and coded from scratch in PyTorch (custom convolutional backbone, feature extraction blocks, and dual output heads for bounding box regression and multi-class classification).

Tracking Component

Multi-object tracking (benchmarking Centroid Tracking vs. ByteTrack) to maintain consistent IDs for workers across video frames over time.

Libraries & Modes

Libraries: PyTorch (`torch`, `torch.nn`), Torchvision, OpenCV (`cv2`), NumPy.

Inference Modes: Live webcam frame-by-frame processing and static image upload evaluation.

SPEAKER SCRIPT (~45 SEC)

"Rather than relying entirely on an off-the-shelf YOLO library, I am manually building the PyTorch detector architecture from scratch to handle coordinate regression and classification. OpenCV handles frame capture, and a multi-object tracker maintains persistent worker IDs across video streams."

DATASET SPECIFICATIONS

Roboflow PPE Dataset & 5 Target Classes

SOURCE WORKSPACE

Roboflow Workspace: w211444371-student-hccs-edu

DATASET VOLUME

2,750 Annotated Site Images

Hardhat	Safety Vest	Mask	Gloves	NO-Gloves
Head protection	High-vis gear	Face shield	Hand protection	Violation label

Data Ingestion & Format: Ingested into Google Colab via Roboflow API / PyTorch DataLoader pipelines with image resizing, channel normalization, and bounding-box coordinate formatting.

SPEAKER SCRIPT (~40 SEC)

"My dataset consists of 2,750 images hosted on Roboflow, focusing on five critical PPE categories: Hardhat, Safety Vest, Mask, Gloves, and an explicit violation class for NO-Gloves. The images and annotations are pulled directly into our training pipeline with standard resizing and normalization."

EVALUATION FRAMEWORK

Accuracy, Frame Rate & Latency Benchmarks

PRIMARY METRIC (DETECTION ACCURACY)

≥ 75%

Mean Average Precision (mAP@0.50)

Evaluated on the unseen test split across all 5 target PPE categories.

SECONDARY METRIC (INFERENCE SPEED)

20–30 FPS

Live Video Latency ≤ 50 ms / frame

Uploaded single-image inference ≤ 1.0 second.

SPEAKER SCRIPT (~35 SEC)

"For accuracy, my target is 75% mAP at IoU 0.50 on the test split, which provides a strong benchmark for a custom model. For speed, my target is 20 to 30 frames per second to ensure fluid, real-time video execution."

EXECUTION ROADMAP

16-Week Project Schedule & Milestones

Week	Phase	Milestone Deliverable
Week 10	🧭 **Blueprint**	Project proposal, architecture design, and repository setup
Week 12	🔌 **First Working Demo**	PyTorch CNN architecture coded; successful forward pass verified
Week 13	🛠 **Make It Yours**	Model trained on Roboflow PPE data; OpenCV live video & tracker linked
Week 14	📈 **Improve & Measure**	Evaluate test mAP@0.50 and FPS; tune loss weighting; refine tracking
Week 15	🎥 **Package & Present**	Record 3–5 min video demo, finalize documentation, and submit final build

🎙 **SPEAKER SCRIPT (~35 SEC)**

"My project follows the 16-week timeline: blueprint submission now at Week 10, first working prototype with the PyTorch architecture by Week 12, full training and tracking integration by Week 13, tuning and metric evaluation in Week 14, and the final demo presentation in Week 15."

RISK MITIGATION & BUDGET

Contingency Plans (Plan B) & Zero-Cost Infrastructure

Risk 1: Outdated/Low Quality

If image quality or bounding boxes prove inconsistent, swap to an alternative verified PPE dataset on Roboflow.

Risk 2: Local File Conversion

If parsing local annotation files breaks the pipeline, stream & download dataset directly via Roboflow API into Google Colab.

Risk 3: Tracking / Zone Logic

If implementing custom restricted zones proves overly complex, postpone zone logic to focus on robust worker ID tracking.

RESOURCES & COMPUTE STACK

Local HP Laptop • Google Colab (Free T4 GPU) • Roboflow API • GitHub

TOTAL BUDGET

$0.00 (100% Free)

SPEAKER SCRIPT (~35 SEC)

"My primary risk is annotation quality in the dataset; if needed, I'll switch to an alternative Roboflow PPE set. To avoid file conversion bugs, I'll download directly via the Roboflow API. If spatial boundary zones become complex, I'll prioritize core worker tracking first. All tools used are 100% free."